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The Spatiotemporal Weighted Efficient Drought Index—A new generalized procedure of regional drought indicator

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Abstract

Drought is a multifaceted climate phenomenon. Due to recent climate warming, the risk of drought has increased. Regional drought management requires accurate regional drought characterization to prepare early warning drought mitigation policies. Therefore, a more flexible and efficient procedure is necessary for characterizing drought at a regional level. This study provides a new spatiotemporal weighting scheme for integrating precipitation data from various meteorological stations located in a specific region. Consequently, this article presents a new generalized regional drought indicator—The Spatiotemporal Weighted Efficient Drought Index (STWEDI). The methodology of STWEDI consists of two stages: (1) The first stage combines the meteorological data of various stations located in a certain region under the proposed weighting scheme, and (2) the second stage standardizes the cumulative distribution function (CDF) of mixture probability models. In application, this research estimated the time series data of STWEDI of seven regions containing a varying number of meteorological stations. To assess the performance of STWEDI, we compared the time series data of STWEDI with the most commonly used Standardized Precipitation Index (SPI) using the Pearson correlation. Results show that STWEDI is strongly correlated with individual SPIs in all the clusters and scales. Moreover, we found that STWEDI is strongly associated with SPI at all time scales. These statistical measures show that STWEDI is the more effective drought indicator for regional drought monitoring.

KEYWORDS

climate warming, drought, mixture probability models, Standardized Precipitation Index

1 | INTRODUCTION

Growing climate warming is one of the main consequences of drought (Zandalinas et al., 2021). Drought is one of the most devastating natural hazards. Unlike other natural phenomena, drought is recurrently occurring globally. Depending on the geographical characteristics, spatial extent, duration, and drought intensity vary from region to region (Zhou et al., 2020). Drought occurs due to a prolonged absence of rainfall and a shortage of water supply either on the surface or

groundwater. After the industrial revolution, the frequency and severity of drought have increased worldwide (Spinoni et al., 2018).

Drought has prolonged and significant adverse impacts on various aspects of human life. For example, it badly affects agricultural structure (Thompson et al., 2020; Wu et al., 2020), ecosystem and the hydrological energy process (Eslamian et al., 2017). Drought has a significant negative impact on socio-economic development and the natural environment. Due to drought hazards, many countries lose millions of dollars every year. Droughts have caused the most

significant damage in China from 1949 to 1995, and many damages have resulted in economic losses of more than U.S. \$12 billion (Xu et al., 2015). In Pakistan, an extreme agro-meteorological drought was experienced in 2000, which reduced the wheat and maize yields to -54.6% and -29.9% , respectively (Qaiser et al., 2021).

Consequently, droughts are of prominent concern in the outlining and control of water resources. However, continuous evaluation of this hazard and effective monitoring strategies in advance stages may reduce the bad impact of drought. Therefore, several meteorological and hydrological drought monitoring tools have been developed to quantify droughts. Some popular and recent devolved indicator includes Palmer Drought Severity Index (PDSI) (Palmer, 1965), Standardized Precipitation Index (SPI) (McKee et al., 1993), Reconnaissance Drought Index (RDI) (Tsakiris et al., 2007) Standardized Precipitation Evapotranspiration Index (SPEI) (Vicente-Serrano et al., 2010), Joint Deficit Index (JDI) (Kao & Govindaraju, 2010), Standardized Precipitation Temperature Index (SPTI) (Ali et al., 2017), Multivariate Standardized Drought Index (MSDI) (Hao and AghaKouchak, 2013) and Multivariate Standardized Precipitation Index (MSPI) (Bazrafshan et al. (2014).

Furthermore, Svoboda and Fuchs (2016) have systematically and comprehensively enlisted popular drought indices, their uses, characteristics and corresponding hydro-meteorological parameters. However, due to different climatic geographies, subjectivities in the selection of probability models and the varying input parameters, some sort of inaccuracy and uncertainty in drought monitoring and classification is always existed (Stagge et al., 2015). Therefore, studies must explore new algorithms and strategies for precise and more representative characterization of drought, especially at a regional level.

The key feature of climate research is to learn climatological characteristics at the regional level. Drought monitoring and assessment at the regional level is one of the main aspects of regional hydrological studies. In past research, numerous authors have provided various methods for regional drought monitoring. For example, Ali et al. (2020) developed Seasonally Combinative Regional Drought Indicator (SCRDI) under the Bayesian Network approach. In SCRDI, time-series data of SPTI index of various meteorological stations located in homogenous regions are combined systematically. Under the same approach, Raza et al. (2021) and Qamar et al. (2021) developed regional drought monitoring indicators to assess various types of drought. The main limitation of using a Bayesian network is configurations of parameters and time-consuming computation. Similarly, other indices have some constraints and limited scope.

However, a homogenous climatic region having various gauge stations may cause several difficulties in analysis and re-analysis. For example, the inappropriate presence of meteorological stations scattered over the region without any complete drought monitoring framework can mislead the results. For instance, the spatial characterization of drought in one area can be wet while it can be dry in the surrounding. Thus, a compact regional drought indicator is essential for regional drought monitoring and forecasting.

This study aimed to provide a generalized and efficient regional drought monitoring indicator by integrating multiple time series data

of various meteorological stations located in a specific region. This study has two distinct features: (1) to provide a new spatiotemporal weighting scheme for integrating spatiotemporal data of precipitation and (2) to provide mixture probability models for increasing the efficiency of the standardized drought index.

2 | THE PROPOSED INDICATOR FOR REGIONAL DROUGHT: THE SPATIOTEMPORAL WEIGHTED EFFICIENT DROUGHT INDEX (STWEDI)

Here, we provide the methodology of the proposed STWEDI framework. The methodology has been carried out in two phases. The first phase is related to the propagation of the weighting scheme. In this phase, the time series of precipitation is amalgamated from various meteorological stations scattered over the homogenous climatic region. The rationale of the weighting scheme has been comprehensively discussed in the subsection. The second phase is related to the modelling and standardization of the combined time series of precipitation. We have applied K-Component Gaussian Mixture Distribution (K-CGMD) to model weighted combined precipitation time series. These two phases are explained in the subsequent section accordingly.

2.1 | Phase 1: The proposed weighting scheme for integrating regional data

One of the main objectives of this study is to combine homogenous meteorological stations. This section proposes a new weighting scheme for integrating precipitation time series data of various meteorological stations located in specific homogenous clusters/regions. Here, weights are assigned using the probability proportional to size technique. This helps to avoid underrepresenting one subgroup in the study and yields more accurate results. The main feature of the proposed scheme is that the assignment of weights for each meteorological observatory is characterized by data spatially and temporally, independently.

The mathematical expression for the spatiotemporal weighting scheme is defined below.

Let $\tilde{R} \in (S_1, S_2, \dots, S_K)$ is a network of homogenous meteorological stations located in a certain geographical region. To combine time series data of precipitation, the proposed weights are defined as:

Let $X_1, X_2, \dots, X_{nk}$ be the time series of precipitation. To assign the weight spatiotemporally, the precipitation time series of the homogenous climatic region is arranged according to the following mathematical form for a given month. Similarly, the data structure can be generalized for other months too. Let w_{itm} be the temporal weight for i th observation in $\tilde{R}$ with subject to the condition that $\sum_i^n w_{itm} = 1$, where the formula of for calculating w_{itm} depends on the temporal amount and individual contribution in a particular month. Mathematically,

$$w_{itm} = \frac{x_{itm}}{\sum_{i=1}^n x_{itm}}. \quad (1)$$

Let v_{itm} be the spatio-temporal weight for both i th and k th observations in $\tilde{R}$.

The formula of v_{itm} is as follows:

$$v_{itm} = \frac{w_{itm}}{\sum_{t=1}^l w_{itm}}. \quad (2)$$

Here,

$$\sum_i^n v_{itm} = 1. \quad (3)$$

Finally, the spatio-temporal data of precipitation are combined as follows:

$$P_i^c = J\theta(u_{i1}, u_{i2}, \dots, u_{im}) \quad (4)$$

Here,

$$u_{im} = \sum_{t=1}^l x_{itm} \times v_{itm}. \quad (5)$$

The next phase provides analogy and statistical mechanics to standardized P_i^c .

2.2 | Phase 2: K-CGMD-based probabilistic standardization

Fitting a probability model on precipitation time series is mandatory to obtain SDI. Furthermore, fitting a univariate model is common in literature and has been applied in several studies for the calculation of SDI. In the beginning, different univariate models have been extensively applied for modelling the precipitation time series to compute SPI, SPEI, and other drought indexes. The first version of SPI, gamma distribution, was introduced as an appropriate probability model (Edwards, 1997). Guttman (1999) model the precipitation time series using different probability models and concluded that the Pearson Type-III distribution is the best.

Similarly, Vicente-Serrano et al. (2010) used three parameters of log-logistic distribution to model the D_i time series to obtain SPEI. Recently Ali, Hussain, Faisal, Almanjahie et al. (2019) and Jiang et al. (2020) have fitted 36 different probability models using the 'R' package (Spiess, 2014) for the computation of different drought indices. However, it is quite possible to have more than a peak of precipitation time series. Fitting a single probability model on a precipitation data having more than one peak or mode can reduce the accuracy of drought monitoring. The potential meteorological

variable understudy in the evaluation phase of SDI time series data may follow mixture distribution or have multi models distribution. Therefore, it makes intuitive sense to model multimodal data as a mixture of many unimodal Gaussian distributions. Therefore, we have applied K-CGMD to attain more accuracy. K-CGMD contains very interesting theoretical and computational properties; it can fit all the possible distributions on a single iteration (Khan et al., 2022; Li et al., 2022).

The probability density function (pdf) and the cumulative density function (CDF) of K-CGMD with parameters $\theta_k = \{\mu_k, \Sigma_k\}$ are given below.

$$p(x \sim \mu_k, \Sigma_k) = \frac{1}{\sqrt{(2\pi)^d \det(\Sigma_k)}} \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)\right) \quad (6)$$

$$F(x) = \Phi(x) = \int_{-\infty}^x p(x \sim \mu_k, \Sigma_k) dx \quad (7)$$

Since we have monthly precipitation data, it is obvious to choose $K = 12$; the 12-component possible combined CDF can be of the form as follows:

$$F(x) = F(P_{i1}^c) + F(P_{i2}^c) + F(P_{i3}^c) \dots F(P_{i12}^c) \quad (8)$$

This CDF is standardized using the proper standardization procedure in the next section for the calculation of SPI and the proposed drought index (STWEDI).

The SPI and STWEDI values can be obtained by standardizing the CDF of the best-fitted model. This study has normalized the CDF of the K-CGMD given in Equation 10 using classical transformation. Following McKee et al. (1993) and Vicente-Serrano et al. (2010), the standardization procedure for the proposed indicator is as follows.

$$STWEDI = -\left(m + \frac{c_0 + c_1 m + c_2 m^2}{1 + d_1 m + d_2 m^2 + d_3 m^3}\right) \quad (9)$$

For

$$m = \sqrt{\ln\left(\frac{1}{\{F(x)\}^2}\right)} \quad (10)$$

When

$$0 \leq F(x) \leq 0.5 \quad (11)$$

Where $F(x)$ is the CDF of 12-CGMD.

And,

$$STWEDI = +\left(m - \frac{c_0 + c_1 m + c_2 m^2}{1 + d_1 m + d_2 m^2 + d_3 m^3}\right) \quad (12)$$

Where

$$m = \sqrt{\ln\left(\frac{1}{\{1 - F(x)\}^2}\right)} \quad (13)$$

When

$$0.5 \leq F(x) \leq 1 \quad (14)$$

$c_0 = 2.515517$, $c_1 = 0.802853$, $c_2 = 0.010328$, $d_1 = 1.432788$, $d_2 = 0.985269$, and $d_3 = 0.001308$ are constants.

This study proposed STWEDI based on the spatiotemporally weighted combined time-series (STWCT) of precipitation time series. The STWEDI drought index used the STWCT instead of using station-based/grid-based usual precipitation time series. STWEDI is a regional-based and more efficient and more reliable drought index. Therefore, we have used the STWCT for obtaining the STWEDI. Moreover, drought has several categories. Table 1 shows the classification of drought categories of wet and dryness of STWEDI.

3 | COMPARATIVE EVALUATION AND QUALITY MEASURE

SPI is a popular and commonly used index for drought monitoring and assessment. This index is developed by McKee et al. (1993). SPI is a probabilistic index, and it has the characteristic of multi-time scaling. The computation method of SPI is quite simple and easy. It is computed by fitting the precipitation time series to the probability model and transforming it into a normal distribution to obtain a standardized value. The modelling and standardization procedure for the computation of STWEDI is presented in the next section. The same procedure has been adopted for SPI as well.

In meteorology, ecology, and hydrology literature, numerous studies used the Pearson correlation coefficient (r) for their comparative analysis. Wang et al. (2017) used a correlation coefficient

for the comparison and evolutionary characteristics of three drought indices. Similarly, Myrionidis et al. (2018) have compared 10 notable drought indices through correlation coefficient. Therefore, this study has also compared the proposed drought index with the popular SPI drought index using the Pearson correlation coefficient. The Pearson correlation coefficient is commonly denoted by r and is a statistical measurement used to find the linear relationship between two arrays or variables. The range of r is always between -1 and 1 . A positive value close to 1 indicates a strong positive correlation, while a negative value near -1 shows a strong inverse relationship. Furthermore, the zero value indicates no linear relationship.

4 | APPLICATION

Pakistan is one of the top 10 countries that has been significantly affected by climate change (Jamro et al., 2019). In the last decades, the climate of Pakistan has severely changed, especially since the variability of the monsoon is increased. The possible influencing factor of climate change is global warming, the receding of Himalayan glaciers, and the frequent occurrence of drought hazards. As a result, the country's capacity for water reservoirs and hydropower is reduced. The higher atmospheric and sea temperature and low precipitation could also add to salt stress. Moreover, the frequent occurrence of drought episodes has badly affected the whole ecosystem, agriculture, and livestock of the country.

Consequently, drought has become an essential problem for the ecosystem, agriculture and human lives. Several studies have focused on tackling the major problem of drought and providing many methods for monitoring and quantifying drought in Pakistan. For example, Adnan et al. (2018) used various drought indices to monitor and assess the drought status. Similarly, Ali et al. (2020) have suggested an ensemble procedure for regional drought pattern recognition.

However, accurate assessment and characterization of drought are still challenging. In this study, an attempt has been made to propose a method for accurate and efficient drought monitoring. Here, we have computed the SPI and the proposed drought index STWEDI by utilizing the precipitation (mm) time series of seven homogeneous clusters for the period 1970–2016. These seven homogeneous clusters continued 52 meteorological stations. These homogeneous clusters are designed by Ali, Hussain, Faisal, Shoukry et al. (2019). The time-series data contained monthly average precipitation. The precipitation time series data have been collected by Pakistan meteorological department, Islamabad, Pakistan. These 52 meteorological stations are located across the whole region of Pakistan. Figure 1 shows the location of these selected stations, where the summary statistics of the clusters are presented in Table 2. The summary statistics contained the first quartile, second quartile and mean.

TABLE 1 Drought classification criterion

Range of STWEDI and SPI	Drought classification
2.00 and above	Extremely wet
1.50 to 1.99	very wet
1.00 to 1.49	Moderate wet
−0.99 to 0.99	Near normal
−1.00 to −1.49	Moderate drought
−1.50 to −1.99	Severe drought
−2.00 and less	Extremely drought

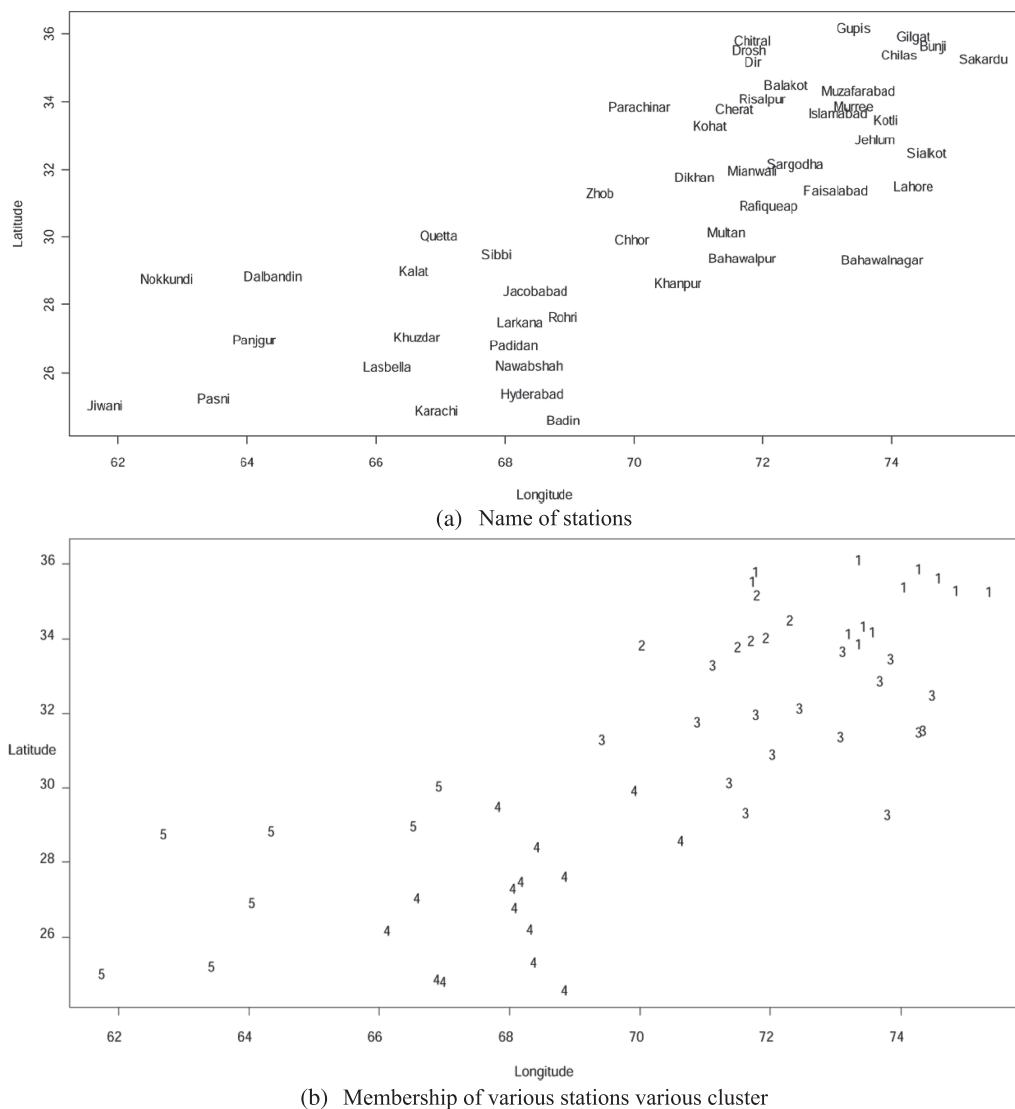


FIGURE 1 Maps of the metrological stations showing the membership in respective clusters. (a) Name of stations. (b) Membership of various stations various cluster

5 | RESULTS AND DISCUSSION

5.1 | Spatiotemporal integration of precipitation data in the regional context under the proposed weighting scheme

This section presents experimental results associated with integrating precipitation data under the proposed weighting scheme. In this study, the spatiotemporal weights are assigned to monthly precipitation time series using the probability proportional to size technique. By Equations 1 and 2, these weights are calculated for each month corresponding to the stations. These weights reflect two data dimensions during the integration process (i.e., spatially and temporally). After the estimation of weights, time-series data of

various meteorological stations are combined accordingly. In the same way, the spatiotemporal weighted time series of precipitation is obtained for all seven homogenous regions. The observed values of precipitation are selected based on weights. The spatiotemporally weighted time series of precipitation from various months and stations are then combined. Finally, the STWCT of precipitation is modelled for the drought index. The STWCT is regional precipitation. Figure 2 shows the temporal plots of spatiotemporal weights assigned to the observed time series of cluster 1. Similarly, Figure 3 shows the monthly weighted temporal plots of only Astore stations. These plots show the fluctuations of weights assigned to observed precipitation. Moreover, Table 3 provides the individual weighted values of cluster 1 for all months.

TABLE 2 Cluster-wise summary statistics of precipitation data

Months	Description statistics	Clusters						
		Cluster 1 11	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7
Jan	Q1	0.30	12.20	33.20	1.00	0.00	1.10	0.01
	Mean	16.43	46.25	88.13	15.52	3.22	22.04	19.84
	Q3	21.43	62.00	126.35	21.50	3.60	31.15	22.80
Feb	Q1	2.00	30.20	69.49	5.30	0.00	0.95	0.00
	Mean	20.14	79.09	128.98	25.61	6.72	23.96	14.67
	Q3	30.75	106.20	171.75	33.10	8.65	34.50	13.40
Mar	Q1	6.08	41.90	75.50	4.60	0.00	0.75	0.00
	Mean	30.74	106.80	145.31	31.89	6.41	24.27	12.32
	Q3	42.20	139.10	203.65	46.30	7.05	36.05	12.00
Apr	Q1	8.68	26.90	43.70	3.20	0.00	0.00	0.00
	Mean	38.73	78.61	98.68	22.54	4.27	10.54	3.00
	Q3	51.70	113.90	140.85	31.00	3.80	13.30	0.00
May	Q1	7.78	14.10	34.35	2.00	0.00	0.00	0.00
	Mean	33.61	44.29	65.85	16.34	3.42	6.47	0.30
	Q3	46.20	62.30	89.20	23.00	1.75	7.05	0.00
June	Q1	2.60	6.50	56.35	6.00	0.00	0.00	0.00
	Mean	13.92	28.34	110.96	38.63	8.10	6.56	7.82
	Q3	18.70	39.40	142.10	56.10	6.00	3.50	0.15
July	Q1	5.25	19.50	195.45	41.40	0.01	0.00	0.00
	Mean	16.25	81.36	293.98	122.30	42.94	21.49	24.95
	Q3	22.03	117.10	368.25	178.10	59.70	23.30	19.00
Aug	Q1	5.80	21.30	182.15	26.30	0.06	0.00	0.00
	Mean	20.20	88.19	269.98	112.28	48.39	18.38	24.95
	Q3	27.23	129.00	337.50	146.00	70.40	21.00	23.70
Sep	Q1	2.68	10.00	60.45	4.60	0.00	0.00	0.00
	Mean	14.53	40.44	116.99	47.18	20.83	4.05	6.82
	Q3	15.53	60.20	149.70	65.00	12.00	0.00	0.01
Oct	Q1	0.00	2.80	10.20	0.00	0.00	0.00	0.00
	Mean	9.42	27.88	46.00	9.38	2.95	3.41	1.76
	Q3	8.90	36.00	65.40	10.20	0.00	0.00	0.00
Nov	Q1	0.00	0.10	2.50	0.00	0.00	0.00	0.00
	Mean	6.27	21.39	29.96	3.96	1.20	3.29	1.89
	Q3	6.13	26.70	44.10	3.60	0.00	1.75	0.00
Dec	Q1	0.00	2.00	7.90	0.00	0.00	0.00	0.00
	Mean	11.23	28.69	53.36	8.34	4.02	14.34	14.29
	Q3	12.40	40.00	72.90	8.40	1.00	14.35	13.00

5.2 | Estimations of SPI and STWEDI

In this study, monthly average precipitation time series ranging from 1970 to 2016 for seven homogeneous regions/clusters containing 52 meteorological stations scattered across the whole region of Pakistan are used to compute SPI. Similarly, the proposed spatiotemporally weighted precipitation time-series (STWPT) is used to

calculate STWEDI at various time scales. The STWEDI drought index is based on STWPT. We have discussed the computational rationale of STWEDI in the methodology section. The standardization section of STWEDI is based on the principle rationale of McKee et al. (1993).

Moreover, we have calculated the SPI drought index to check, validate, and compare the candidacy of STWEDI. Both of these

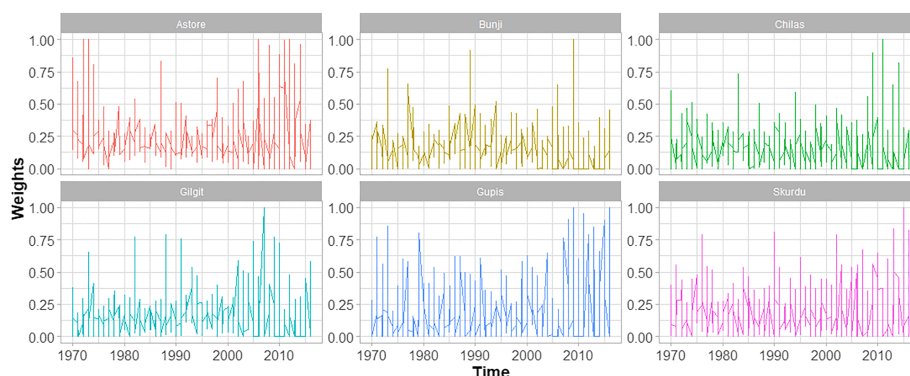


FIGURE 2 Temporal behaviour of the proposed weights for individual stations in cluster 1

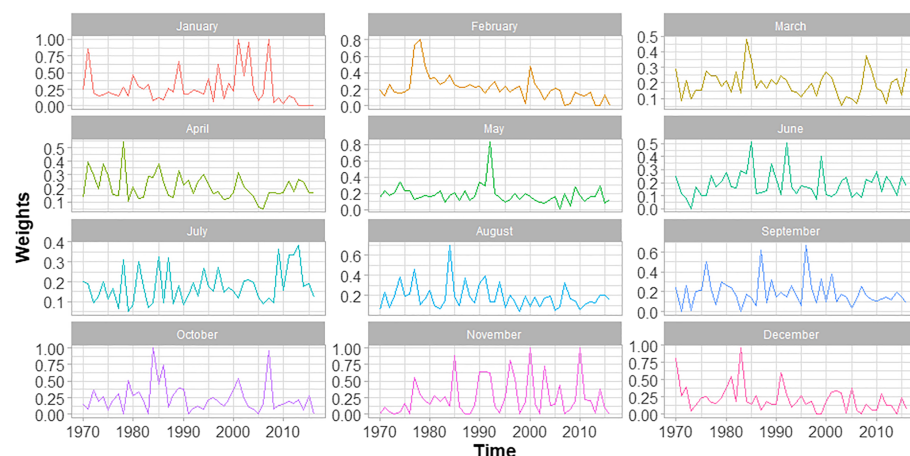


FIGURE 3 Monthly temporal behaviour of the proposed weights at Astore station

indices have been calculated for all the time scales (i.e., 1, 3, 6, 9, 12, 24 and 48). The simple precipitation and the spatiotemporal weighted precipitation are modelled using K-CGMD for SPI and STWEDI, respectively. Figure 4 shows the K-CGMD density plots and Q-Q scatter plots of STWEDI for the seven clusters at time scale 1. The Q-Q plots indicate that the fitted model is the best one as the probability model has covered almost all precipitation time series.

Tables 4 and 5 provide the BIC values of the 12-CGMD for STWEDI and SPI, respectively. In Table 4, the BIC values of STWEDI for cluster 1 at scale 1, scale 3, scale 6, scale 9, scale 12, scale 24 and scale 48 are -4604.707 , -5616.025 , -6052.615 , -6191.113 , -6302.498 , -6587.522 and -6497.800 , respectively. The rest of the BIC values for other clusters can be found in Table 4. Similarly, the BIC values of SPI of separate stations can be observed in Table 5.

Figure 5 shows the temporal plots of all the time scales of SPI and STWEDI for cluster 7. The cluster 7 contains three meteorological stations, that is, Jiwani, Karachi and Pasni. Here, we have perceived mostly similar scatter behaviour for both indices.

5.3 | Comparative assessment using Pearson correlation coefficient (r)

The most important features of regional indicators are consistency and strong association with the observatories. This section compares and discusses the association of STWEDI with SPI. The comparative analyses are based on the Pearson correlation coefficient (r). Table 6 shows the correlation between STWEDI and SPI on various time scales. Since this study is based on the seven homogeneous clusters of 52 stations, STWEDI has been compared with SPI data of individual meteorological stations. The correlation of STWEDI with SPI at a 1-month time scale in Astore, Bunji, Chilas, Gilgit, Gupis, and Skardu (cluster 1) are 0.8791, 0.7287, 0.7514, 0.7273 and 0.5999, respectively. These high values of correlation reveal that STWEDI is a strong candidate for analysing drought at the regional level. Similarly, high correlations have been observed in 3-, 6-, 9-, 12-, 24- and 48-month time scales. This analysis implies that the use of STWEDI for regional drought monitoring is rationally valid.

TABLE 3 Spatiotemporal weights for cluster 1

Stations	Jan.	Feb.	Mar.	Apr.	May.	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
Astore	0.6566	0.22092	0.19210	0.21302	0.18486	0.18770	0.17795	0.18731	0.18930	0.24288	0.25910	0.22251
Bunji	0.153294	0.129411	0.166912	0.161244	0.160667	0.182176	0.16638	0.160996	0.151985	0.135765	0.090316	0.101042
Chilas	0.15329	0.12941	0.16691	0.16124	0.16067	0.18218	0.16638	0.16100	0.15199	0.13577	0.09032	0.10104
Gilgit	0.125175	0.133264	0.148827	0.156182	0.161497	0.170147	0.171602	0.160726	0.177998	0.157298	0.122711	0.110169
Gupis	0.12786	0.17200	0.15923	0.16838	0.15259	0.14635	0.15388	0.15670	0.14357	0.09689	0.13266	0.11819
Sakardu	0.173499	0.220366	0.167744	0.159909	0.148787	0.170419	0.161214	0.168846	0.151821	0.145231	0.146276	0.22434
Sum of weights	1	1	1	1	1	1	1	1	1	1	1	1

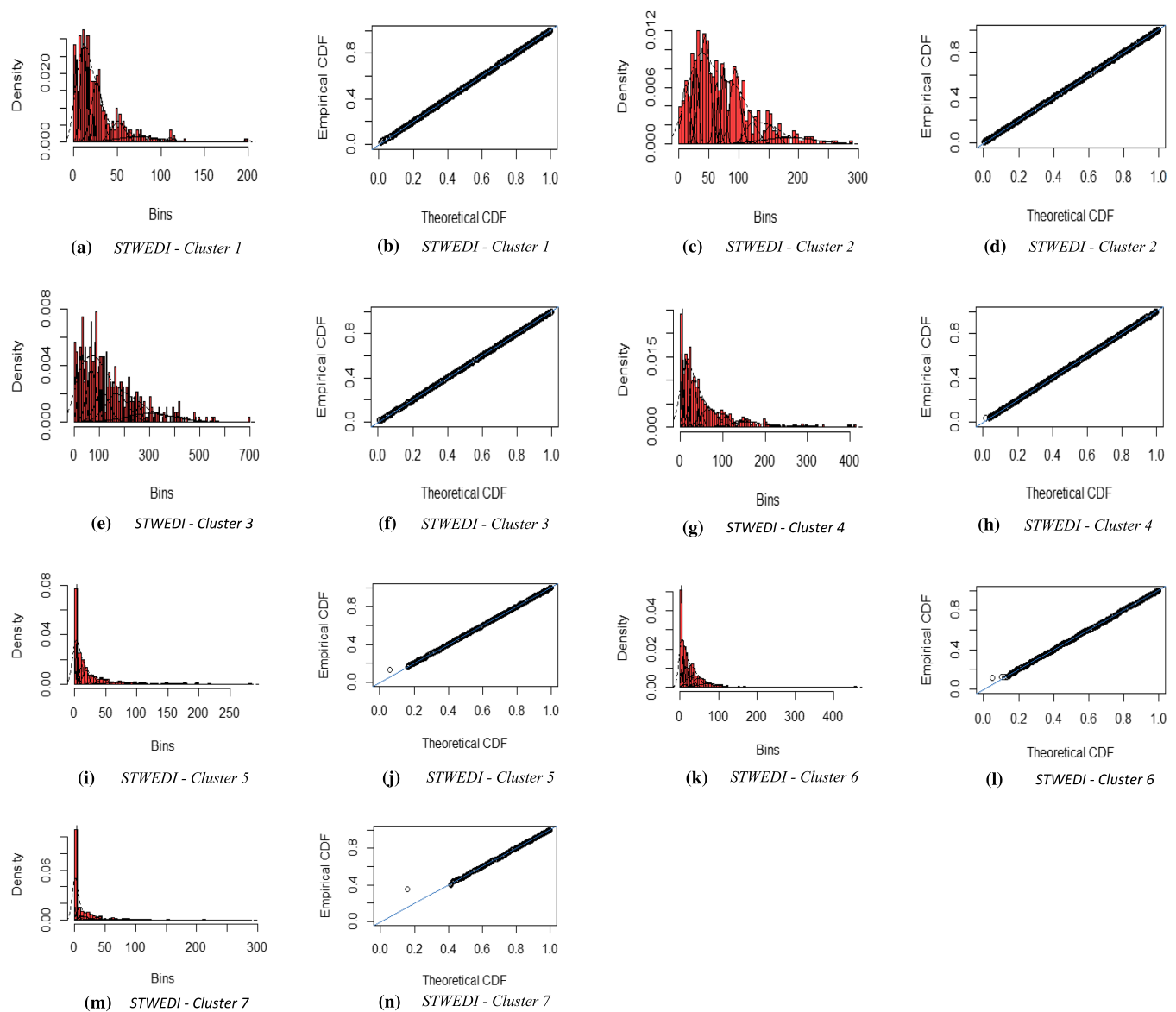


FIGURE 4 Mixture distribution and Q-Q plots of STWEDI at one mount time scale

TABLE 4 BIC values of K-CGMD of SWTEDI in all clusters at various time scales

Clusters	Time scales						
	1	3	6	9	12	24	48
1	-4604.707	-5616.025	-6052.615	-6191.113	-6302.498	-6587.522	-6497.800
2	-5657.617	-6472.805	-6748.871	-6907.072	-6954.595	-7220.529	-7275.243
3	-6458.648	-7381.632	-7572.629	-7614.813	-7405.517	-7663.187	-7832.305
4	-5277.431	-6586.843	-7038.154	-7072.161	-6858.884	-7067.325	-7044.622
5	-3658.165	-5788.494	-6458.604	-6645.325	-6694.726	-6869.374	-6980.697
6	-2902.786	-5696.217	-6024.318	-6267.697	-6357.180	-6602.136	-6792.035
7	-1611.697	-5060.772	-6540.002	-6193.733	-6632.183	-6946.373	-7137.167

TABLE 5 BIC values of K-CGMD for SPI at various stations and time scales participating in the cluster

Clusters	Stations	Time scales						
		1	3	6	9	12	24	48
Cluster 1	Astore	−6226.455	−6203.161	−6664.967	−6737.222	−6791.656	−7039.224	−7032.156
	Bunji	−2969.899	−5006.138	−5593.140	−5772.658	−5873.561	−6115.885	−6038.685
	Chilas	−3206.720	−5279.562	−5875.445	−6116.417	−6246.007	−6563.623	−6657.499
	Gilgit	−3047.011	−4929.688	−5492.996	−5615.464	−5676.318	−5932.842	−5926.377
	Gupis	−4663.160	−5345.756	−5974.483	−6305.287	−6521.169	−6903.006	−7221.063
	Skardu	−3681.413	−5473.306	−6000.671	−6227.099	−6338.572	−6598.685	−6665.302
Cluster 2	Cherat	−4867.766	−6507.232	−6867.943	−7070.973	−7116.184	−7288.710	−7185.683
	Chitral	−4487.986	−6204.958	−6782.096	−6790.024	−6724.139	−7011.670	−7018.643
	Dir	−6251.380	−7132.541	−7394.232	−7564.138	−7585.098	−7895.159	−7985.863
	Drosh	−5282.080	−6344.424	−6796.681	−6844.860	−6778.073	−6984.308	−6991.200
	Kohat	−5015.706	−6421.923	−6801.012	−7032.274	−7100.905	−7418.546	−7428.822
	Parachinar	−5898.504	−6938.213	−7347.775	−7516.179	−7540.933	−7895.268	−7966.748
	Peshawar	−4643.060	−6241.225	−6638.569	−6880.069	−6968.834	−7289.481	−7381.561
	Risalpur	−5101.089	−6613.208	−6938.648	−7163.349	−7093.986	−7194.216	−7268.194
	Zhob	−4930.120	−5732.127	−6150.600	−6318.061	−6330.710	−6635.839	−6727.069
Cluster 3	Balakot	−6466.709	−7415.352	−7639.863	−7778.972	−8633.557	−7957.771	−8146.594
	Garhidupatta	−6373.192	−7285.705	−7505.861	−7644.736	−7611.946	−7808.974	−7795.542
	Islamabad	−5906.225	−7311.598	−7738.733	−7845.424	−7686.002	−7935.776	−7829.035
	Kakul	−6243.620	−7189.805	−7394.212	−7422.049	−7246.189	−7489.969	−7575.935
	Kotli	−5882.863	−7272.504	−7571.593	−7678.834	−7537.230	−7647.463	−7634.652
	Murree	−6551.851	−7543.867	−7776.465	−7849.932	−7787.762	−8086.057	−8097.973
	Muzaffarabad	−6367.848	−7368.758	−7643.429	−7721.471	−7593.346	−7769.253	−7841.120
Cluster 4	Bahawalnagar	−2404.207	−5356.973	−6246.078	−6457.329	−6527.554	−6924.829	−6978.113
	Bahawalpur	−4065.392	−4527.142	−5869.403	−6273.162	−6377.728	−6615.520	−6558.316
	D.I. Khan	−3609.085	−5777.694	−6339.409	−6506.545	−6531.015	−6817.212	−6919.306
	Faisalabad	−3621.557	−6094.947	−6695.379	−6816.512	−6702.084	−7042.165	−7164.340
	Lahore	−4713.807	−6690.800	−7273.602	−7410.456	−7328.068	−7604.189	−7518.565
	Mianwali	−4466.308	−6492.837	−7005.731	−7165.989	−7172.064	−7390.408	−7441.744
	Multan	−2547.344	−5273.676	−6052.122	−6206.624	−6258.806	−6373.049	−6494.478
	Sargodha	−4383.027	−6317.958	−6822.010	−6879.578	−6791.381	−7071.376	−7057.509
	Sialkot	−5331.895	−7119.830	−7701.288	−7832.817	−7649.151	−7757.521	−7786.420
Cluster 5	Badin	−2736.663	−3264.798	−5616.560	−6536.612	−6511.041	−6842.208	−6944.646
	Chhor	−3225.235	−3523.596	−5744.671	−6501.210	−6595.771	−6768.994	−6691.404
	Hyderabad	−3125.688	−3559.166	−5499.568	−6091.762	−6332.828	−6693.775	−6647.663
	Jacobabad	−2436.180	−3915.272	−5463.201	−5886.933	−6206.734	−6552.499	−6707.263
	Khanpur	−2623.882	−3940.161	−5537.654	−5905.247	−6114.718	−6489.400	−6512.341
	Larkana	−1235.716	−3505.172	−5256.861	−5688.006	−5928.094	−6399.142	−6588.068
	Mohenjodaro	−1709.726	−2822.817	−5167.710	−5548.175	−5747.587	−6162.744	−6073.306
	Nawabshah	−2834.969	−3358.614	−5496.555	−6032.479	−6299.017	−6689.113	−6879.113
	Padidan	−1836.739	−3356.234	−5310.507	−5836.948	−6084.067	−6544.604	−6814.245
	Rohri	−1861.334	−3196.621	−4991.039	−5761.082	−5985.876	−6400.476	−6548.473
	Sibi	−1507.988	−2980.424	−5856.364	−6145.901	−6292.739	−6562.677	−6630.559
Cluster 6	Dalbandin	−1229.555	−1421.102	−2862.956	−5396.552	−5564.552	−5730.973	−5854.565
	Kalat	−2152.316	−2496.041	−5921.791	−6349.312	−6547.101	−7015.974	−7121.739
	Khuzdar	−1635.563	−5629.252	−6138.930	−6404.827	−6522.094	−6683.862	−6843.566
	Lasbela	−1800.301	−2633.095	−5926.835	−6286.823	−6388.683	−6541.995	−6715.149
	Nokkundi	−1073.910	−1124.051	−2076.652	−3086.439	−4690.689	−5182.748	−5449.764

TABLE 5 (Continued)

Clusters	Stations	Time scales						
		1	3	6	9	12	24	48
Cluster 7	Panjgur	-1355.429	-3740.619	-5233.540	-5618.302	-5724.713	-6013.249	
	Quetta	-2242.834	-4810.770	-6320.454	-6537.732	-6546.279	-6847.260	-7035.556
	Jiwani	-1194.355	-1010.924	-2939.600	-5721.069	-5903.444	-6238.536	-6221.524
	Karachi	-83.170	-3903.557	-5800.477	-6373.379	-6512.048	-6934.631	-7049.840
	Pasni	-173.215	-1096.631	-3593.954	-5698.559	-5922.544	-6273.802	-6327.212

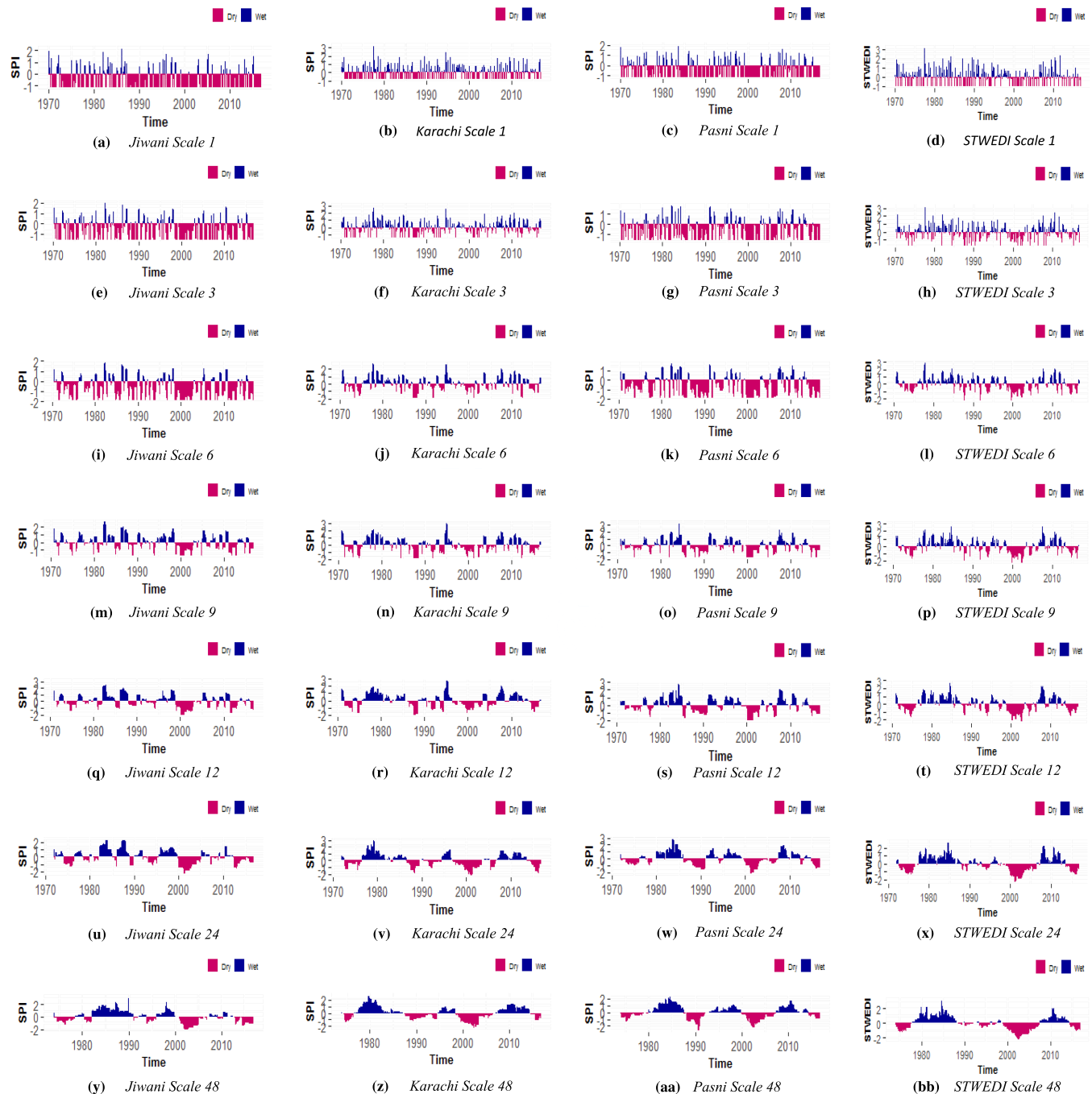


FIGURE 5 Temporal behaviour of SPI at various station in cluster 7 and STWED

TABLE 6 Correlation coefficient (r) between STWEDI and SPI in individual stations at each cluster

Clusters	Stations	Time scales						
		1	3	6	9	12	24	48
Cluster 1	Astore	0.8791	0.8864	0.8831	0.8503	0.8292	0.8010	0.7153
	Bunji	0.7287	0.7319	0.6933	0.6715	0.6447	0.6032	0.5192
	Chilas	0.7514	0.7967	0.7673	0.7075	0.6607	0.6456	0.6733
	Gilgit	0.7273	0.7172	0.6989	0.6799	0.6566	0.6402	0.5090
	Gupis	0.5999	0.6270	0.6226	0.6260	0.6367	0.6687	0.7355
	Skardu	0.7826	0.7841	0.7684	0.7360	0.7006	0.6609	0.5263
Cluster 2	Cherat	0.7723	0.7773	0.6845	0.6563	0.5938	0.5980	0.8052
	Chitral	0.4210	0.3065	0.3723	0.4184	0.4009	0.2782	0.1740
	Dir	0.8538	0.7792	0.6365	0.6144	0.5072	0.3535	0.2536
	Drosh	0.5369	0.4351	0.4249	0.4055	0.3706	0.2699	0.1031
	Kohat	0.7381	0.7251	0.5808	0.5381	0.4053	0.3175	0.2952
	Parachinar	0.7669	0.8217	0.7807	0.7709	0.7566	0.8248	0.8400
	Peshawar	0.7768	0.7837	0.7026	0.6975	0.6600	0.6570	0.7385
	Risalpur	0.7906	0.7716	0.6729	0.6760	0.6451	0.6192	0.7155
	Zhob	0.5809	0.5988	0.4477	0.3184	0.1416	0.0538	0.1028
Cluster 3	Balakot	0.9313	0.9343	0.8934	0.8600	0.7930	0.8051	0.8494
	Garhidupatta	0.9010	0.8723	0.7787	0.7505	0.6709	0.6559	0.7174
	Islamabad	0.8916	0.8853	0.8121	0.7143	0.5392	0.4464	0.3950
	Kakul	0.9379	0.9446	0.9262	0.8921	0.8289	0.8418	0.9011
	Kotli	0.9035	0.9239	0.8873	0.8201	0.6919	0.6511	0.7375
	Murree	0.9400	0.9423	0.9102	0.8809	0.8136	0.8031	0.8408
	Muzaffarabad	0.9234	0.9136	0.8742	0.8434	0.7613	0.7279	0.7569
Cluster 4	Bahawalnagar	0.6540	0.7295	0.7084	0.6022	0.4693	0.5146	0.4842
	Bahawalpur	0.6063	0.6935	0.6761	0.6000	0.5532	0.4732	0.4559
	D.I. Khan	0.7030	0.7411	0.6660	0.5563	0.4215	0.4360	0.5015
	Faisalabad	0.8023	0.8475	0.8205	0.7199	0.5027	0.4712	0.3668
	Lahore	0.8904	0.9211	0.9302	0.9060	0.8428	0.8381	0.8414
	Mianwali	0.7845	0.8093	0.7522	0.6625	0.5506	0.5555	0.6493
	Multan	0.6531	0.7280	0.7115	0.6444	0.5616	0.5615	0.6494
	Sargodha	0.8340	0.8819	0.8500	0.7560	0.6022	0.5165	0.3348
	Sialkot	0.8983	0.9149	0.9176	0.8737	0.7741	0.6678	0.5401
Cluster 5	Badin	0.7013	0.7916	0.8328	0.8465	0.8304	0.8269	0.8231
	Chhor	0.7563	0.8318	0.8720	0.8792	0.8485	0.8266	0.7984
	Hyderabad	0.6957	0.7620	0.7772	0.7367	0.6743	0.5346	0.4197
	Jacobabad	0.6329	0.6581	0.6838	0.7216	0.7259	0.7205	0.8179
	Khanpur	0.6668	0.7071	0.6955	0.6309	0.5482	0.4014	0.4413
	Larkana	0.4377	0.4032	0.3970	0.4335	0.4352	0.3910	0.4312
	Mohenjodaro	0.2182	0.2728	0.1904	0.0870	-0.0861	0.0437	0.2746
	Nawabshah	0.7009	0.4533	0.3714	0.2449	0.0765	0.5499	0.7907
	Padidan	0.6636	0.7377	0.7743	0.7782	0.7478	0.7599	0.8249
	Rohri	0.6234	0.6561	0.6646	0.6564	0.6450	0.6485	0.7525
	Sibi	0.6906	0.6914	0.6327	0.6137	0.5914	0.6641	0.8035
Cluster 6	Dalbandin	0.4784	0.4885	0.5017	0.5757	0.5774	0.5449	0.5830
	Kalat	0.0472	0.0424	0.5540	0.5849	0.5484	0.4967	0.5168
	Khuzdar	0.7427	0.7485	0.6800	0.7235	0.7345	0.7014	0.6934

TABLE 6 (Continued)

Clusters	Stations	Time scales						
		1	3	6	9	12	24	48
Cluster 7	Lasbela	0.5612	0.6331	0.6275	0.6362	0.6150	0.5894	
	Nokkundi	0.4039	0.4044	0.3464	0.4183	0.3455	0.3006	0.2745
	Panjgur	0.5756	0.6418	0.6349	0.6812	0.6579	0.5913	0.5909
	Quetta	0.6320	0.5930	0.5792	0.6884	0.7541	0.7178	0.5712
	Jiwani	0.5737	0.5209	0.4864	0.5592	0.5535	0.5550	0.6727
	Karachi	0.8060	0.7446	0.6708	0.7402	0.7461	0.7774	0.7577
	Pasni	0.6087	0.5436	0.5058	0.6126	0.6505	0.7301	0.7706

6 | CONCLUSION

This study proposed a new generalized regional drought monitoring indicator by integrating the precipitation data of various meteorological stations located in a specific territory/region. The integration of precipitation data is based on the newly proposed spatiotemporal weighting scheme. At the same time, the standardization process of the integrated information is based on K-CGMD. In application, this research considered seven different climatic regions of Pakistan. Each region has varying numbers of homogeneous climate stations. Based on the correlation analysis, the comparative results show that the proposed regional indicator strongly correlates with SPI data of individual meteorological stations in respective regions.

From this study, we conclude the following remarks:

1. For effective drought mitigation policies at the regional level, accurate integrated meteorological data are required to estimate SDIs type drought indicators.
2. Integrating multivariate precipitation data under the proposed weighting scheme is rationally valid.
3. Under the mixture probability model, the accuracy in the estimated data of SDIs can be enhanced.

The proposed indicator integrates precipitation data only. This limits the proposal to assess drought under a single climate variable. Therefore, a generalized procedure that uses other meteorological variables such as temperature, solar radiation and humidity can be propagated to enhance the effectiveness of regional drought monitoring.

CONFLICT OF INTEREST

The authors declare no competing interests.

AUTHOR CONTRIBUTIONS

All authors have an equal contribution.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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